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# HydraDG: Governed Context Interventions with Fractal Custody for Agent Experiments

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## Abstract

1 Agent evaluations increasingly depend on context interventions—retrieval, mem-  
2 ory graphs, and prompt contracts—yet published reports often omit null outcomes,  
3 parser failures, and negative cells. We present HydraDG, a governed experimen-  
4 tal framework that binds substantive model invocations, tool calls, and handoffs to  
5 *Fractal Custody Objects* (FCOs) and a *Fractal Custody Graph* (FCG). HydraDG  
6 preregisters hypotheses, freezes case manifests and scorers, records raw prompt  
7 and response hashes, and preserves terminal states including timeouts, abstentions,  
8 and malformed outputs without silent substitution. We report two preregistered  
9 studies (EXP-008, EXP-009) comparing flat prose versus structured FCG retrieval  
10 for failure-prevention endpoints; both terminate as *underpowered* under bounded  
11 case replication, so confirmatory effect claims are not supported while negative  
12 and null evidence remains in custody. We argue that custody-first evaluation in-  
13 frastructure supports reproducible agent science rather than optional logging.

## 14 1 Introduction

15 Large language model agents are evaluated on task success, benchmark scores, and human prefer-  
16 ence [2, 3]. Less often preserved are the conditions under which experiments fail: swapped models,  
17 stale runtime selection, partial matrix execution, or JSON outputs that never parse. When negative  
18 and null cells disappear from the public record, later work cannot distinguish *no effect* from *no*  
19 *measurement*.

20 HydraDG addresses this gap with a custody-first experimental contract. Every actor—human oper-  
21 ator, frontier agent, local model runtime, or deterministic script—must emit or be represented by a  
22 handoff receipt before its output is promoted. Receipts bind actor identity, host, repository state, in-  
23 put hashes, raw output hashes, evidence class, claim ceiling, and signature state (hashes are identity  
24 commitments, not digital signatures unless an authorized signing operation occurred). Probabilis-  
25 tic model outputs are never treated as authority; they are evidence objects subject to deterministic  
26 scoring and graph append rules.

27 This paper describes the HydraDG framework and reports terminal results from the IC Failure Learn-  
28 ing structured-retrieval falsification lane. We do *not* claim that structured FCG context improves  
29 failure prevention; preregistered EXP-008 and EXP-009 closed as underpowered. We also do not  
30 treat exploratory secondary patterns as promoted findings.

## 31 2 Related Work

32 **Agent evaluation.** Benchmark suites measure capability but rarely require publication of failed  
33 runs, malformed generations, or infrastructure blocks [2, 3]. HydraDG complements benchmarks  
34 with mandatory negative capture at the custody layer.

35 **Retrieval and structured context.** Retrieval-augmented generation and graph-augmented memory  
36 systems assume structured context helps [1, 4]. Falsification requires frozen contracts, explicit null  
37 retention, and power-aware verdicts; HydraDG enforces both.

38 **Provenance, integrity, and reproducibility.** W3C PROV-O and workflow provenance systems  
39 represent process and research-object lineage [12, 13], while append-only Merkle audit logs pro-  
40 vide established tamper-evident integrity mechanisms [11]; HydraDG does not claim to invent these  
41 primitives. Recent work surveys evidence tracing and execution provenance for LLM agents [15],  
42 and separate work argues for preserving negative ML results rather than selective omission [14].  
43 Preregistration, FAIR principles, and nanopublication-style lineage further support reproducibility  
44 [5–7]. HydraDG integrates rather than replaces these foundations: probabilistic outputs, determin-  
45 istic transforms, malformed cells, failures, abstentions, blocked states, and underpowered termina-  
46 tions remain typed evidence with explicit claim-promotion ceilings enforced via FCO/FCG handoff  
47 receipts. Companion custody-framework preprints (deferred to camera-ready de-anonymization) de-  
48 fine FCO/FCG formally; they establish framework provenance, not the HydraDG empirical results  
49 reported here.

## 50 3 Framework

### 51 3.1 Custody objects

52 An **FCO** (*Fractal Custody Object*) materializes one bounded transformation: e.g., a model call,  
53 scorer pass, or ingest operation. An **FCG** (*Fractal Custody Graph*) append records directed edges  
54 between FCOs with hash-linked roots [11]. HydraDG preserves failures, nulls, and malformed  
55 outputs in custody (a failure-complete *behavior*, not a redefinition of FCO/FCG). Claim ceilings  
56 (exploratory mechanistic falsification, descriptive only, etc.) are declared at preregistration and en-  
57 forced at promotion time.

### 58 3.2 Experimental freeze

59 Each experiment declares:

- 60 • frozen case manifest and prompt-contract hashes;
- 61 • conditions (e.g., flat prose vs. structured FCG);
- 62 • models with expected runtime digests;
- 63 • primary endpoint and aggregation rule;
- 64 • explicit gates for confirmatory vs. exploratory claims.

65 Execution emits `START_RECEIPT`, per-cell raw outputs with prompt/response SHA-256, parser state,  
66 and terminal receipts. Integrity failure (duplicate keys, missing dependencies, silent model substi-  
67 tution) blocks promotion; scientific failure (timeout, abstention, malformed JSON) is retained as  
68 terminal evidence.

### 69 3.3 Governed local execution

70 Where preregistered, local model calls route through a governed bridge that records selection age,  
71 swap posture, and resolved digest. Direct runtime invocation and governed invocation are separate  
72 evidence classes; one is not silently labeled as the other.

Table 1: Terminal preregistered studies (HydraDG IC Failure Learning). *Underpowered* denotes insufficient paired evidence for confirmatory promotion, not proof of null effect.

| Study   | Primary verdict | Raw cells | Valid parse rate |
|---------|-----------------|-----------|------------------|
| EXP-008 | UNDERPOWERED    | 300       | 0.907            |
| EXP-009 | UNDERPOWERED    | 300       | 0.883            |

## 73 4 Experimental Setup

### 74 4.1 Task family

75 The IC Failure Learning suite uses synthetic agent-failure vignettes with preregistered endpoints  
76 such as E06 (prevents exposure of out-of-vault media). Cases are drawn from a frozen CASES .jsonl  
77 manifest; each case receives multiple replicates per condition.

### 78 4.2 Conditions and models

79 **C0 (FLAT\_PROSE)**: failure-handling guidance as unstructured text.

80 **C1 (STRUCTURED\_FCG)**: the same semantic content composed through a frozen FCG retrieval  
81 contract.

82 EXP-008 and EXP-009 used local models qwen3:1.7b and qwen2.5-coder:7b under identical  
83 case and scorer contracts. Thinking/reasoning modes were held consistent with preregistration. Pri-  
84 mary aggregation: majority of three replicates per case.

### 85 4.3 AI and agent methodology disclosure

86 Frontier coding agents assisted with repository tooling and manuscript preparation; they are not  
87 listed as authors. All scientific endpoints come from preregistered local model executions with  
88 hashed prompts and responses. Routine editing does not constitute a reported experimental interven-  
89 tion.

### 90 4.4 Power and claim gates

91 E06 endpoints are known limited-power under the frozen case set. Confirmatory ordering claims  
92 require preregistered discordant pairs; exploratory secondary statistics are quarantined from primary  
93 promotion.

## 94 5 Results

95 Table 1 summarizes terminal verdicts for preregistered studies. Both experiments executed the full  
96 raw matrix ( $n = 300$  cells each) with complete custody append.

97 **EXP-008**. Structured FCG retrieval did not yield promotable confirmatory evidence on E06 under  
98 the frozen design ( $n = 2$  paired cases per model stratum at aggregation scope). Non-zero malformed  
99 and abstention rates are preserved in custody.

100 **EXP-009**. Primary verdict remains UNDERPOWERED for confirmatory E06 ordering. An ex-  
101 ploratory secondary pattern was recorded as directionally positive at the descriptive layer only; it is  
102 **not promoted** to a causal claim.

103 **Stage-2 baseline**. Prior IC Failure Learning stage-2 execution (414 scored rows across two models)  
104 established FAILURE\_LEARNING\_BEHAVIOR\_IMPROVEMENT\_NOT\_ESTABLISHED, providing histor-  
105 ical context for the structured-retrieval falsification lane.

### 106 5.1 Failure-preserving systems validation

107 The preregistered retrieval experiments above test whether structured context improves a frozen end-  
108 point. A separate systems-validation suite asks a different question: does the custody mechanism

Table 2: Systems-validation outcomes (custody mechanics only; not treatment-effect evidence).

| Validation              | Scope     | Outcome                            |
|-------------------------|-----------|------------------------------------|
| Perturbation matrix     | 100 cells | 100/100 chain verification         |
| Synthetic tamper suite  | 8 modes   | 8/8 detected                       |
| Concurrent execution    | 10 runs   | PASS (10 unique run IDs)           |
| Replay/restart recovery | 44 events | PASS                               |
| Live provider ladder    | R0–R6     | Bounded external failure preserved |

109 preserve integrity under perturbation, tampering, concurrency, replay, and external execution fail-  
 110 ure? Table 2 summarizes source-verified outcomes from a frozen HydraLamp perturbation matrix;  
 111 these results evaluate custody mechanics and do **not** constitute evidence that structured FCG context  
 112 improves the EXP-008/009 endpoint.

113 The tamper suite used synthetic cases only (not real security incidents). In a live provider repair lad-  
 114 der, earlier infrastructure gates passed before a structured call hit an external quota limit; the failure  
 115 remained a terminal auditable state rather than being erased or substituted. The same custody primi-  
 116 tives have been instantiated in sibling systems for local-model orchestration, deterministic scientific-  
 117 document atomization, offline anomaly detection, bounded bioinformatics, substrate generation, and  
 118 operator-facing scientific workflow monitoring; these implementations motivate transfer studies but  
 119 do not provide additional evidence for the treatment effect reported here.

## 120 6 Discussion

121 HydraDG shows that agent experiments can retain null and failure cells with the negative-result in-  
 122 tegrity expected in preregistered experimental sciences: failures are first-class, hashes are mandatory,  
 123 and claim ceilings block over-interpretation. The EXP-008/009 underpowered terminations bound  
 124 what cannot yet be claimed while preserving malformed and timeout cells for audit.

### 125 6.1 Limitations

126 Limitations include bounded case replication, local-only model lanes in the reported matrix, and  
 127 absence of runtime-equivalent successor replication in primary results. A preregistered Qwen 3.8  
 128 successor lane exists but is non-terminal and omitted from primary results. Whole-project hierarchi-  
 129 cal atomization (SeedGraph v1a validation) ran for multiple days but was interrupted during final  
 130 Parquet serialization without a BUILD\_RECEIPT; partial node/edge artifacts are not readback-safe,  
 131 so terminal whole-corpus atomization is not established in this submission. Biological, protein-  
 132 structure, and cellular perturbation applications of the custody framework remain future validation  
 133 targets, not demonstrated generalizations of the agent-context results here.

### 134 6.2 Future directions

135 **Agent systems.** Larger successor replications, governed multi-agent handoffs, selective replay, and  
 136 resource-aware custody gates under preregistered manifests.

137 **Structured retrieval at scale.** Checkpointed ingest and transactionally durable atomization so long  
 138 hierarchy builds can resume from batch boundaries rather than materializing the entire graph at  
 139 finalization.

140 **Cross-project federation.** Project-local FCG roots with explicit cross-project references and no  
 141 automatic evidence admission between repositories.

142 **Cryptographic custody.** Authorized signatures and MMR commitments where policy permits;  
 143 hashes remain identity commitments unless a verified signing operation occurred.

144 All directions above are *planned* unless independently measured; they do not extend the EXP-  
 145 008/009 empirical claims.

## 7 Conclusion

We presented HydraDG, a custody-first framework for agent context-intervention experiments, and reported two terminal underpowered preregistered studies that block premature promotion of structured FCG retrieval effects on failure-prevention endpoints. We recommend FCO/FCG custody, explicit claim ceilings, and terminal-state preservation as baseline infrastructure for NewInML-style agent evaluations.

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Table 3: Terminal preregistered study audit (appendix).

| Study   | Verdict      | Raw cells | Valid parse rate |
|---------|--------------|-----------|------------------|
| EXP-008 | UNDERPOWERED | 300       | 0.907            |
| EXP-009 | UNDERPOWERED | 300       | 0.883            |

## A Full EXP-008 / EXP-009 statistical audit

Table 3 records terminal verdicts with parse rates reverse-traced to frozen verdict JSON receipts.  $p$ -values remain *not informative* under the preregistered underpowered design.

## B Stage-2 M0/M1/M2 baseline

Stage-2 IC Failure Learning execution (414 scored rows across two models) closed as FAILURE\_LEARNING\_BEHAVIOR\_IMPROVEMENT\_NOT\_ESTABLISHED; descriptive only.

## C HydraLamp systems-validation matrix

See Table 4 for custody-mechanics outcomes (not treatment-effect evidence).

Table 4: Systems-validation matrix (appendix).

| Validation              | Scope     | Outcome                            |
|-------------------------|-----------|------------------------------------|
| Perturbation matrix     | 100 cells | 100/100 chain verification         |
| Synthetic tamper suite  | 8 modes   | 8/8 detected                       |
| Concurrent execution    | 10 runs   | PASS (10 unique run IDs)           |
| Replay/restart recovery | 44 events | PASS                               |
| Live provider ladder    | R0–R6     | Bounded external failure preserved |

## D Deterministic verification and tooling

Figure 1 and DETERMINISTIC\_AUDIT\_TOOLING\_LEDGER.jsonl document the custody verification pipeline with frozen script SHA-256 and R1/R2/R3 roots.

Source bytes → SHA-256 → Atom → ResultAtom → Claim/Table → FCG delta → exact-SHA verification

Figure 1: FIG-001: governed custody pipeline from source bytes to exact-SHA verification (structured data; self-generated).

## E Citation and reference custody

Portfolio recovery bounded 185 occurrences to unique identities; only proposition-backed references are admitted. Citation fabrication literature informs desk-rejection risk [16], [17], [13], [18], [12].

## F Requirement, template, and deadline drift

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205 **G Prior-art comparison matrix**

206 Related-work boundaries include PROV-O, CWLProv, and Workflow Run RO-Crate [16], [17], [13],  
207 [18], [12]. Prior-art search responses remain *DISCOVERY\_ONLY*; not promoted to verified empiri-  
208 cal novelty proof.

209 **H SeedGraph / FCO / FCG custody audit**

210 TOTAL\_VERIFIED\_INGEST\_COMPLETE=NO with verified coverage 31.55% (307/973).

211 **I Planned and nonterminal lanes**

212 GPU SGLang canary, Q38 XENV, and full verified ingest remain nonterminal. All appendix cita-  
213 tions resolve through the single main-text bibliography authority.

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